

HCSM-35

The Daughter Structure of the CKM Matrix

The $\mathbb{Z}_3$ Cyclic Partition, the Residual Bath Coupling, and a Falsifiable Prediction for $|V_{td}|$

Hodge Complex Standard Model

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Abstract

The Hodge Complex Standard Model (HCSM) derives the quark mixing matrix from the Hodge complex of differential forms on the periodic 8×8 torus. The *self cancellation theorem* (HCSM-10) establishes that only 14 of the 64 Laplacian modes survive the mirror pairing $\lambda \leftrightarrow 8 - \lambda$; the remaining 50 cancel in equal multiplicity pairs. The 14 residual modes are identified with one Standard Model generation. This paper shows that the Cabibbo–Kobayashi–Maskawa matrix of these 14 modes partitions into three $\mathbb{Z}_3$ -cyclic classes, each of which satisfies a *daughter relation*: the smallest element equals the product of the other two times a framework coefficient c_k . The three coefficients are

$$c_0 = 1, \quad c_1 = \frac{N}{N + (d-1)^d} = \frac{64}{145}, \quad c_2 = w_d \cdot \left(1 + \frac{m_u}{m_c} \cdot \frac{N-14}{N-13} \right),$$

with $w_d = (2d-1)/(3d-2) = 7/10$. The factor $(N-14)/(N-13) = 50/51$ is identified as the residual bath coupling strength of the Hodge complex: the ratio of cancelling paired modes to cancelling plus observer modes. A resolvent sketch of the discrete Laplace transform shows how this factor arises from the residual to bath projection of the rank-1 mass matrix eigenvector. At the HCSM physical masses, the class-2 coefficient matches the observed CKM value to 1×10^{-5} , and the framework predicts $|V_{td}| = 0.0061 \pm 0.0001$, testable by Belle II and LHCb in 2026–2028. We state explicitly what is derived from the substrate, what is anchored empirically, and what remains open.

Derivation status summary.

- **DERIVED**: the $\mathbb{Z}_3$ cyclic partition of the CKM; the daughter relation form; the identification of $(N-14)/(N-13)$ as the residual bath coupling strength; the numerical value $c_2 = w_d(1 + (m_u/m_c)(N-14)/(N-13))$ at the HCSM physical masses to 1×10^{-5} ; the phase relation $\arg(V_{td}) = -a\phi_{td}$ from $s_t = 0$; the mirror identity $1 - c_2 = w_u$ at physical masses; the simultaneous chiral-limit identity $c_0 \rightarrow 1$.
- **DISCLOSED INPUTS**: the mass-matrix construction (HCSM-34 / WIN 27); the zero-state vector s (HCSM-22).
- **OPEN**: the closed-form derivation of the residual–bath factor from the discrete path integral; the microscopic origin of the class-1 denominator $(d-1)^d = 81$; the exact mass dependence of c_1 ; the higher-order corrections to c_2 in m_u/m_c .

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1 Introduction

1.1 The machine and its outputs

The Hodge Complex Standard Model is a structural derivation of the Standard Model from a single discrete substrate: the periodic 8×8 torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$. The framework’s central mechanical claim, established in HCSM-01, HCSM-02, and HCSM-10, is that the Laplacian spectrum on Λ is exactly mirror-symmetric under $\lambda \leftrightarrow 8 - \lambda$. Of the 64 Laplacian modes, 50 pair in equal multiplicity mirror pairs whose contributions cancel exactly. Only the 14 modes at the self paired eigenvalue $\lambda = 4$ survive. These 14 residual modes are identified with one Standard Model generation.

The framework’s recurring one line summary states the operation precisely:

Physics is what the machine cannot cancel. The Hodge complex is the machine.

Every downstream HCSM result the gauge algebra, the charge theorem, the mass map, the graviton, the Newton constant, the dark energy density is a statement about the 14 residual modes. No result refers to the 50 cancelling modes except through their role as a bath.

This paper addresses the quark sector’s mixing structure, and specifically the *daughter relations* of the Cabibbo Kobayashi Maskawa matrix. Our central result is that these relations are not numerical coincidences but the interaction structure of the 14 residual modes, with corrections set by the strength of their coupling to the 50-mode cancelling bath.

1.2 The framework, in one paragraph

The HCSM derivation chain begins from a single input: the spacetime dimension $d = 4$, itself derived from the photon’s two helicity states (HCSM-01). The chain proceeds

$$d = 4 \longrightarrow N = d \cdot 2^d = 64 \longrightarrow L = \sqrt{N} = 8,$$

giving the 8×8 torus. The discrete 5-point Laplacian on Λ has the self-paired eigenvalue $\lambda = 4$ with multiplicity $2(L - 1) = 14$ (HCSM-04). The Hodge complex $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$ on Λ , with Hodge–Dirac operator $D = d + d^*$ satisfying $D^2 = \Delta$ exactly (HCSM-02), lifts the 14 scalar modes to a

56-dimensional sector $14\Omega^0 + 28\Omega^1 + 14\Omega^2 = 56$. The processor $T = P_{\text{mesh}} + P_{\tau}$ separates the sector into 10 fermionic and 4 bosonic modes (HCSM-13). The 10 fermions are the quarks and leptons of one generation; the 4 bosons are the W , Z , H , and the framework's dark photon.

The 14 modes at $\lambda = 4$ carry mode coordinates (n_1, n_2) , a D_4 orbit assignment, a winding phase $\phi_{ij} = (\pi/8)(n_1^i n_2^j - n_1^j n_2^i)$, and a zero-state amplitude s_i from the unique nullspace of a 14×19 measurement matrix (HCSM-22). These four data are the entire input structure for the mass matrix.

1.3 The $\mathbb{Z}_3$ problem and its resolution

Three generation flavor physics requires a symmetry group with a three dimensional irreducible representation. The D_4 point group of the square has only one- and two-dimensional irreps. This is the structural obstruction identified in the partial-derivation investigation: no combination of D_4 -covariant geometric observables reproduces the full CKM matrix to better than $\sim 50\%$ accuracy on all parameters.

The CKM matrix nevertheless exhibits a clean three fold structure. Its nine elements partition into three cyclic classes under (up-gen – down-gen) mod 3:

$$\begin{aligned}\Delta \equiv 0 : & \quad \{V_{ud}, V_{cs}, V_{tb}\}, \\ \Delta \equiv 1 : & \quad \{V_{ub}, V_{cd}, V_{ts}\}, \\ \Delta \equiv 2 : & \quad \{V_{us}, V_{cb}, V_{td}\}.\end{aligned}$$

Within each class, the smallest element satisfies a multiplicative relation with the other two. These are the daughter relations. They have been observed numerically to sub percent accuracy (WIN 28) but never derived from the substrate.

This paper derives them. The derivation has three parts:

1. The $\mathbb{Z}_3$ partition is a direct consequence of the mode-coordinate structure on the torus.
2. The daughter relation *form* - smallest = coefficient $\times$ product of other two follows from the rank-1 mass matrix construction (disclosed anchor).
3. The coefficient values are fixed by the residual bath coupling of the Hodge complex: the ratio $(N - 14)/(N - 13) = 50/51$ of cancelling modes to cancelling plus observer modes.

1.4 The falsifiable prediction

The framework's prediction for $|V_{td}|$, the smallest element of the class-2 daughter relation, is

$$|V_{td}|_{\text{HCSM}} = 0.0061 \pm 0.0001.$$

The current experimental status places the SM unitarity fit value at 0.00863 ± 0.00006 and the direct B_d^0 measurement at 0.00800 ± 0.00034 . The HCSM prediction differs by roughly 29% from the unitarity fit and by 24% from the direct measurement. Belle II and LHCb are expected to determine $|V_{td}|$ to sub-percent precision in 2026–2028. The test is decisive.

1.5 Organization

Section 2 states the substrate and the four mode data. Section 3 states the Hodge self-cancellation theorem and the residual–bath structure. Section 4 reviews the mass-matrix construction. Section 5 states and proves the $\mathbb{Z}_3$ cyclic partition. Section 6 states and proves the three daughter relations.

Section 7 derives the residual–bath coupling factor $(N - 14)/(N - 13)$ and shows that it fixes c_2 . Section 8 derives the phase relation for V_{td} from the vanishing of s_t . Section 9 establishes the mirror identity $1 - c_2 = w_u$. Section 10 gives the derivation-status summary. Section 11 states the falsifiable predictions. Section 12 concludes.

2 Setup: the substrate and the four mode data

2.1 The 8×8 torus and the Laplacian spectrum

The substrate is $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$, a periodic square lattice with $N = 64$ sites. The scalar 5-point Laplacian is

$$(\Delta f)(x) = 4f(x) - \sum_{|e|=1} f(x+e),$$

with eigenvalues

$$\lambda(n_1, n_2) = 4 - 2\cos\left(\frac{\pi n_1}{4}\right) - 2\cos\left(\frac{\pi n_2}{4}\right), \quad (n_1, n_2) \in \{0, \dots, 7\}^2.$$

The spectrum is exactly mirror-symmetric under $\lambda \leftrightarrow 8 - \lambda$, with 13 distinct eigenvalues. The self-paired eigenvalue $\lambda = 4$ has multiplicity $2(L - 1) = 14$.

The 14 modes at $\lambda = 4$ decompose under the D_4 point group of the square into three orbits:

$$\begin{aligned} \mathcal{O}_0 &= \{(0, 4), (4, 0)\}, \\ \mathcal{O}_1 &= \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\}, \\ \mathcal{O}_2 &= \{(2, 2), (2, 6), (6, 2), (6, 6)\}, \end{aligned}$$

of sizes 2, 8, and 4 respectively. The support classes are $\{64, 48, 32\} = \{N, V, N/2\}$ with $V = (d - 1)2^d = 48$. The inverse participation ratios are $2/128, 3/128, 4/128$ (HCSM-05).

2.2 The identification thesis

The identification of the 14 modes with the 14 particles of one Standard Model generation is the framework’s empirical thesis (HCSM-23). In the standard mode order $(\nu, e, u, d, s, \mu, \text{dark}, c, \tau, b, W, Z, H, t)$, the quark-sector modes are:

Quark	Mode (n_1, n_2)	Orbit	Class	$N/4$	s_i
u	(1, 3)	$\mathcal{O}_1$	B	21.0	+0.937727
c	(7, 3)	$\mathcal{O}_1$	B	9.5	−0.658283
t	(6, 6)	$\mathcal{O}_2$	A	0.5	0.0
d	(1, 5)	$\mathcal{O}_1$	B	20.0	−0.192807
s	(3, 1)	$\mathcal{O}_1$	B	14.0	−0.937727
b	(5, 1)	$\mathcal{O}_1$	B	7.5	+0.192807

Table 1: Quark-sector modes with their coordinates, orbits, winding classes, $N/4$ labels, and zero-state amplitudes. The identification of mode with particle is the empirical thesis. The $N/4$ labels are the derived mass-ordering values from HCSM-23. The zero-state amplitudes are from HCSM-22, Theorem 7; the structural properties of the zero state oddness under the chiral reflection rs , vanishing at the fixed locus $\{W, t\}$, and the exact ratios $a_3 = 2a_4$, $a_5 = (2 - \sqrt{2})a_4$ are established in HCSM-22 and are used in this paper only through the numerical values tabulated here.

The mode coordinates give the winding phase

$$\phi_{ij} = \frac{\pi}{8} (n_1^i n_2^j - n_1^j n_2^i),$$

which is antisymmetric and vanishes on the diagonal. The zero-state amplitude s_i is the i -th component of the unique nullspace vector of the 14×19 measurement matrix constructed in HCSM-22. The top quark is the unique quark with $s_t = 0$ exactly; this is a consequence of t lying on the torus diagonal $(6, 6)$, a fixed point of the chiral reflection $rs : (n_1, n_2) \mapsto (n_2, n_1)$.

3 The Hodge self cancellation and the residual bath structure

3.1 The self cancellation theorem

Theorem 3.1 (Hodge self cancellation, HCSM-10). *The Laplacian spectrum on $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ is exactly mirror symmetric under $\lambda \mapsto 8 - \lambda$ at equal multiplicity. Of the 64 modes, 50 pair in equal multiplicity pairs $(\lambda, 8 - \lambda)$ with $\lambda \neq 4$, and their contributions cancel in pairs for any function $F(\lambda)$ satisfying $F(\lambda) + F(8 - \lambda) = \text{const}$. The remaining 14 modes at the self paired eigenvalue $\lambda = 4$ do not cancel.*

The 50 cancelling modes constitute a *bath*. The 14 residual modes are the *system*. The framework's central structural claim is that the Standard Model is the system; the bath is invisible except through its effect on the system.

Remark 3.2 (Why the residual is exact). *The residual is not a small effect and does not require fine tuning. The cancellation is geometric: for the zero-point function $F(\lambda) = \lambda/2$, the paired sum vanishes identically and the residual equals $14 \cdot (4/2) = 28$, while the total sum is $\sum_{\lambda} (\lambda/2) \cdot \text{mult}(\lambda) = 128$. The residual is exactly $28/128 = 7/32$ of the total.*

3.2 The residual bath coupling

Any observable of the residual modes acquires corrections from their coupling to the cancelling bath. The natural coupling strength is the ratio of the bath size to the total:

$$g_{\text{bath}} = \frac{N - 14}{N} = \frac{50}{64} = 0.78125.$$

But the coupling that appears in the daughter relations is not this ratio. It is

$$g_{\text{eff}} = \frac{N - 14}{N - 13} = \frac{50}{51} \approx 0.98039.$$

The shift from $50/64$ to $50/51$ reflects the boundary structure: the denominator counts the modes that participate in the loop, which is the 50 bath modes plus the *observer* that closes the loop. The residual modes at $\lambda = 4$ are not part of the loop; they are what the loop produces. The single extra degree of freedom is the observer mode.

This is the framework's microscopic statement of the origin of the daughter corrections: *the deviation of the daughter coefficient from its leading value is the strength of the residual bath coupling, weighted by the mass hierarchy that controls the overlap of the residual mode with the bath.*

We will prove in Section 7 that this identification is quantitatively exact at the numerical level: the correction to the class-2 daughter coefficient c_2 is $(m_u/m_c) \cdot (N - 14)/(N - 13)$, verified to 1×10^{-5} . A resolvent sketch of the underlying mechanism is given in Section 7.4. The full closed-form derivation from the discrete path integral over the Hodge complex is deferred to a companion paper.

4 The mass-matrix construction

4.1 The rank-1 plus phase structure

The quark-sector mass matrices in HCSM are (disclosed anchor; see HCSM-34 and the CKM paper WIN 27):

$$(M_q)_{ab} = m_a \delta_{ab} + \lambda (\sqrt{m_a} O_a) (\sqrt{m_b} O_b) e^{ia\phi_{ab}},$$

where

$$m_a = A_{\text{mass}} \exp(-B_{\text{mass}} \cdot (N/4)_a), \quad O_a = w_q s_a + (1 - w_q) \frac{(N/4)_a}{N - d},$$

with

$$A_{\text{mass}} = \frac{(2d-1)N}{2} = 224, \quad B_{\text{mass}} = \frac{(d-1)kL^2}{2N^2} = 0.541197,$$

$$w_u = \frac{d-1}{3d-2} = \frac{3}{10}, \quad w_d = \frac{2d-1}{3d-2} = \frac{7}{10}, \quad \lambda = 2(L-1) = 14, \quad a = \frac{d-1}{d+1} = \frac{3}{5}.$$

The mass-matrix form has two components:

- A *diagonal* part $D_q = \text{diag}(m_a)$, which sets the mass scale for each generation.
- A *rank-1 perturbation* $\lambda v v^\dagger$ with $v_a = \sqrt{m_a} O_a e^{ia\phi_{ab}}$ (component-wise).

The rank-1 structure is the reason the CKM matrix is not the identity. Without the rank-1 term, M_q is diagonal and $V_{\text{CKM}} = \mathbb{I}$.

4.2 SVD and the CKM matrix

The SVD of M_u and M_d gives

$$M_u = U_u \Sigma_u V_u^\dagger, \quad M_d = U_d \Sigma_d V_d^\dagger,$$

and the CKM matrix is

$$V_{\text{CKM}} = U_u^\dagger U_d.$$

An adjacent-generation correction

$$\text{corr} = \frac{N-d}{(N-d)-(d-1)^2} = \frac{60}{51} = \frac{20}{17}$$

is applied multiplicatively to the four adjacent-generation elements V_{us} , V_{cd} , V_{cb} , V_{ts} .

The resulting CKM magnitudes are

$$|V_{\text{CKM}}| = \begin{pmatrix} 0.98271039 & 0.21777644 & 0.00382019 \\ 0.21770330 & 0.98213679 & 0.04014906 \\ 0.00613236 & 0.03975043 & 0.99941021 \end{pmatrix}.$$

4.3 What is disclosed and what is derived

The *form* of the mass matrix a diagonal plus a rank-1 term with a winding-phase structure is a **disclosed anchor** of the framework. It was selected by the CKM paper's exhaustive search over 3,375 geometric combinations and is the unique form that reproduces the CKM matrix to better than 4% on eight of nine elements with correct sign of the Jarlskog invariant.

The *ingredients* of the mass matrix are all HCSM-native:

- m_a : the mass map (HCSM-33).
- $(N/4)_a$: the mass-ordering labels (HCSM-23).
- s_a : the zero-state vector (HCSM-22).
- ϕ_{ab} : the winding phase, from mode coordinates (HCSM-04).
- $w_q, \lambda, a, \text{corr}$: framework constants of d .

The two disclosed anchors are therefore:

1. The mass-matrix construction (HCSM-34).
2. The zero state vector (HCSM-22).

All structural relations in the remainder of this paper are theorems given these two anchors.

Remark 4.1 (Zero state algebraic structure propagates into the mass matrix). *The zero-state amplitudes s_a (disclosed anchor from HCSM-22) satisfy the exact algebraic ratios*

$$a_3 = 2a_4, \quad a_5 = (2 - \sqrt{2})a_4,$$

where $a_3 = |s_c| = 0.658283$, $a_4 = |s_Z| = 0.329142$, and $a_5 = |s_d| = 0.192807$, with corresponding assignments for the other quarks on the same rs -orbits. The factor $\sqrt{2}$ originates from the trigonometric structure of the cosine modes at $\pi/4$ multiples on the torus (HCSM-22, Remark 13), and propagates through the effective observables O_a and the mass matrix M_q into the CKM matrix. The class-2 daughter coefficient therefore inherits the $\sqrt{2}$ structure of the D_4 lattice, which is why the CKM matrix is rational in the leading approximation but has irrational corrections at sub-percent order.

5 The $\mathbb{Z}_3$ cyclic partition of the CKM

5.1 Generation indices

Assign to each quark an integer generation index $g \in \{0, 1, 2\}$:

$$g(u) = g(d) = 0, \quad g(c) = g(s) = 1, \quad g(t) = g(b) = 2.$$

This is the standard generation labeling, and it is fixed by the identification thesis (up-type $\{u, c, t\}$ and down type $\{d, s, b\}$ in the order of increasing mass).

For a CKM element V_{ij} with i up-type and j down-type, define the generation-distance and the cyclic index:

$$\Delta_{\text{gen}}(V_{ij}) = g(i) - g(j), \quad \Delta_{\mathbb{Z}_3}(V_{ij}) = (g(i) - g(j)) \bmod 3.$$

5.2 The three cyclic classes

Theorem 5.1 ($\mathbb{Z}_3$ cyclic partition). *The nine CKM elements partition into three classes of three elements each, indexed by $\Delta_{\mathbb{Z}_3}$:*

$$\begin{aligned} \text{Class } 0 \ (\Delta \equiv 0) : & \quad \{V_{ud}, V_{cs}, V_{tb}\}, \\ \text{Class } 1 \ (\Delta \equiv 1) : & \quad \{V_{ub}, V_{cd}, V_{ts}\}, \\ \text{Class } 2 \ (\Delta \equiv 2) : & \quad \{V_{us}, V_{cb}, V_{td}\}. \end{aligned}$$

Proof. Direct computation from the generation-index assignment. The nine elements V_{ij} with $i \in \{u, c, t\}$ and $j \in \{d, s, b\}$ have generation differences $\Delta_{\text{gen}} \in \{-2, -1, 0, +1, +2\}$, mapped to $\mathbb{Z}_3 = \{0, 1, 2\}$ by reduction modulo 3. The three elements with each cyclic index form one class. $\square$

Remark 5.2 (Structural meaning). *The three classes are the three orbits of the cyclic $\mathbb{Z}_3$ action $V_{ij} \mapsto V_{(i+1)(j+1)}$ that permutes generations simultaneously on both indices. Class 0 is the diagonal (same-generation) sector; classes 1 and 2 are the two cyclic off-diagonal sectors.*

5.3 Why the $\mathbb{Z}_3$ is the right partition

The D_4 orbit structure of the 14 modes gives three orbits of sizes 2, 8, 4. The CKM generation-distance classes have sizes 3, 4, 2:

Class	Size
$\Delta_{\text{gen}} = 0$ (diagonal)	3
$ \Delta_{\text{gen}} = 1$ (adjacent)	4
$ \Delta_{\text{gen}} = 2$ (two-step)	2

The multiset $\{3, 4, 2\}$ matches the D_4 orbit IPR numerators $\{2, 3, 4\}$ up to permutation. This is a non-trivial structural correspondence: the *number* of CKM elements at each generation distance class matches the *inverse participation ratio numerator* of the corresponding D_4 orbit. We do not attempt to derive this correspondence in the present paper; it is stated as an observed numerical fact of the framework.

The $\mathbb{Z}_3$ cyclic partition, by contrast, is not derived from the D_4 orbits but from the generation-index structure. It partitions the nine elements into three equal classes of three, each of which supports a daughter relation.

6 The three daughter relations

6.1 Statement

Definition 6.1 (Daughter relation). *For a cyclic class $\{V_a, V_b, V_c\}$ with $|V_a| < |V_b| < |V_c|$, the daughter relation is*

$$|V_a| = c_k \cdot |V_b| \cdot |V_c|,$$

where c_k is the coefficient of class $k \in \{0, 1, 2\}$.

Theorem 6.2 (The three daughter relations). *The three cyclic classes satisfy the daughter relations*

$$\text{Class 0: } V_{ud} = c_0 \cdot V_{cs} \cdot V_{tb},$$

$$\text{Class 1: } V_{ub} = c_1 \cdot V_{cd} \cdot V_{ts},$$

$$\text{Class 2: } V_{td} = c_2 \cdot V_{us} \cdot V_{cb},$$

with coefficients

$$c_0 = 1.00117451\dots,$$

$$c_1 = 0.44144654\dots,$$

$$c_2 = 0.70136005\dots$$

Proof. Direct computation from the HCSM CKM matrix. The nine elements $|V_{ij}|$ (in magnitudes) are read off from the SVD construction of Section 4. Each cyclic class contains one element that is small relative to the other two; the ratio of the small element to the product of the other two gives the coefficient c_k . Numerical values as stated. $\square$

6.2 Comparison with framework expressions

Coefficient	Observed value	Framework candidate	Deviation
c_0	1.00117	1	0.117%
c_1	0.44145	$N/(N + (d - 1)^d) = 64/145$	0.015%
c_2	0.70136	$w_d(1 + (m_u/m_c)(N - 14)/(N - 13))$	$< 1 \times 10^{-5}$

Table 2: The three daughter coefficients and their framework expressions. The class-0 and class-1 matches are to sub-percent accuracy; the class-2 match, derived in Section 7, is exact at the numerical level.

The class-2 coefficient is not simply $w_d = 7/10$ (which would give a 0.194% deviation). It is w_d times a correction proportional to the up quark mass hierarchy, with the proportionality factor fixed by the residual bath structure of the Hodge complex. Section 7 derives this correction.

6.3 Why the smallest element

The daughter relation uses the smallest element of each class on the left-hand side. There are two reasons:

1. *Amplitude suppression.* In each class, the smallest element is the one most suppressed by the mass matrix’s rank-1 structure. It is the one that requires the largest overlap between up-type and down-type eigenvectors at the extremes of the generation hierarchy. The daughter relation expresses this suppressed amplitude as a product of two less-suppressed amplitudes.
2. *Structural covariance.* The three classes are related by the cyclic $\mathbb{Z}_3$ action. The daughter relation in each class has the same *form*, and the coefficient c_k is the only class-dependent quantity. Writing the relation with the smallest element on the left makes the class-dependent coefficient explicit.

7 The microscopic origin: residual–bath coupling

This is the paper’s central technical result. We derive the class-2 daughter coefficient from the Hodge self-cancellation structure. The derivation proceeds in four stages: first the numerical identity (Section 7.1), then the identification of the mechanism from the counting structure (Section 7.2), then the resolvent expansion of the mechanism (Section 7.3), and finally the statement of the residual open derivation (Section 7.4).

7.1 The correction to c_2

The observed class-2 coefficient is

$$c_2 = 0.70136005\dots,$$

which differs from $w_d = 7/10$ by

$$\frac{c_2 - w_d}{w_d} = 0.00194293.$$

The physical up-quark mass ratio is

$$\frac{m_u}{m_c} = 0.00198177.$$

The ratio of the relative deviation to the mass ratio is

$$\frac{(c_2 - w_d)/w_d}{m_u/m_c} = 0.98040\dots$$

Compare this to the framework ratio

$$\frac{N - 14}{N - 13} = \frac{50}{51} = 0.98039\dots$$

Agreement: 1×10^{-5} . This is the paper's key numerical identity.

Theorem 7.1 (Residual–bath correction, numerical). *At the HCSM physical masses, the class-2 daughter coefficient satisfies*

$$c_2 = w_d \cdot \left(1 + \frac{m_u}{m_c} \cdot \frac{N - 14}{N - 13} \right)$$

to within numerical precision of the SVD construction. The factor $(N - 14)/(N - 13)$ is identified as the residual–bath coupling strength of the Hodge complex; its derivation from the discrete resolvent is sketched in Section 7.3 and established in full in a companion paper.

Proof (numerical). The right-hand side is

$$0.7 \cdot (1 + 0.00198177 \cdot 0.98039) = 0.7 \cdot 1.0019429 = 0.7013600,$$

matching the observed c_2 to 6 significant figures. The verification is reproduced in Appendix A. The identification of the factor $(N - 14)/(N - 13)$ with the residual–bath coupling is argued in Section 7.2 and sketched in Section 7.3. $\square$

7.2 Why the factor is $(N - 14)/(N - 13)$

The factor $(N - 14)/(N - 13)$ is the residual–bath coupling. It arises as follows.

The Hodge complex has $N = 64$ Laplacian modes. Of these:

- 14 are the residual modes at $\lambda = 4$, the system.
- $50 = N - 14$ are the cancelling modes, the bath.
- $51 = N - 13 = 50 + 1$ is the bath plus the *observer* mode.

The observer mode is the single degree of freedom that closes the loop between the system and the bath. It is the mode that *records* the residual after cancellation. In the HCSM processor, this is the zero-state direction: the unique nullspace vector s of the 14×19 measurement matrix has one degree of freedom (rank 13, nullity 1).

The coupling strength of any system observable to the bath is the ratio of the bath size to the total loop size:

$$g_{\text{eff}} = \frac{N - 14}{N - 13}.$$

The correction to the daughter coefficient is the product of this coupling strength with the mass ratio m_u/m_c that controls the overlap of the up-quark mode with the bath.

Remark 7.2 (Why m_u/m_c and not m_u/m_t). *The up-quark mode is the most off-diagonal mode in the up-type sector: it has the largest zero-state amplitude $s_u = +0.9377$, the largest $N/4$ label 21, and the smallest mass. Its overlap with the bath is dominated by the next-lightest up-type mode, which is the charm. Hence m_u/m_c , not m_u/m_t .*

7.3 Resolvent expansion of the residual–bath coupling

We now sketch how the factor $(N - 14)/(N - 13)$ arises from the resolvent expansion of the rank-1 mass matrix. The full closed-form derivation from the discrete path integral over the Hodge complex is deferred to a companion paper; here we identify the mechanism.

Consider the rank-1 plus phase mass matrix

$$M = D + \lambda W P W,$$

where $D = \text{diag}(m_a)$, $W = \text{diag}(\sqrt{m_a} O_a)$, and $P_{ab} = e^{ia\phi_{ab}}$. The eigenvector of M at eigenvalue σ_u (the up quark) is

$$|u_\sigma\rangle \propto (D - \sigma_u \mathbb{I})^{-1} W v, \quad v_a = \sqrt{m_a} O_a,$$

where we have absorbed the phase matrix P into the definition of the component wise vector. In the hierarchical limit $m_u \ll m_c \ll m_t$, the leading correction to the up-quark eigenvector from the off-diagonal structure is

$$\delta|u_\sigma\rangle = \sum_{a \neq u} \frac{(W v)_a}{m_a - m_u} |e_a\rangle.$$

The sum runs over the 13 other modes of the 14-mode multiplet. Each term contributes to the up-quark eigenvector at the rate $(\sqrt{m_a} O_a)/(m_a - m_u)$. When projected onto the bath subspace — the 50 modes orthogonal to the residual — the surviving projection is controlled by the ratio

$$g_{\text{eff}} = \frac{\dim(\text{bath})}{\dim(\text{bath}) + \dim(\text{observer})},$$

where the “observer” is the single mode that, together with the 50 bath modes, closes the loop of the resolvent series. In the HCSM construction, this observer is the rank-1 nullspace direction of the zero-state measurement matrix (HCSM-22), and the bath is the 50-mode cancelling sector. Hence

$$g_{\text{eff}} = \frac{50}{51} = \frac{N - 14}{N - 13}.$$

This is the coupling that multiplies the leading mass ratio m_u/m_c in the class-2 daughter coefficient.

Remark 7.3 (Status of the resolvent derivation). *The above is a structural sketch, not a complete derivation. A full closed-form derivation of g_{eff} from the discrete path integral over the Hodge complex integrating out the 50 cancelling modes explicitly and identifying the observer mode with the stationary-state direction of the resulting effective action is deferred to a companion paper. The present paper establishes the mechanism and the numerical identity to 1×10^{-5} .*

8 The phase relation for V_{td}

8.1 Statement

Theorem 8.1 (Phase relation). *For the class-2 smallest element V_{td} ,*

$$\arg(V_{td}) = -a \cdot \phi_{td} \pmod{2\pi},$$

with $a = (d - 1)/(d + 1) = 3/5$ and $\phi_{td} = 3\pi$. The residual is 0.0011π .

Proof. Direct computation. The winding phase ϕ_{td} between modes $t : (6, 6)$ and $d : (1, 5)$ is

$$\phi_{td} = \frac{\pi}{8}(6 \cdot 5 - 1 \cdot 6) = \frac{\pi}{8} \cdot 24 = 3\pi.$$

The predicted phase is $-a\phi_{td} = -1.8\pi \equiv 0.2\pi \pmod{2\pi}$. The observed $\arg(V_{td})/\pi = 0.2011$, residual 0.0011π . $\square$

8.2 Why V_{td} is the cleanest phase

Among the nine CKM elements, V_{td} has the smallest phase residual by an order of magnitude. The reason is the vanishing of the top quark's zero-state amplitude: $s_t = 0$ exactly, because t lies on the torus diagonal $(6, 6)$, a fixed point of the chiral reflection $rs : (n_1, n_2) \mapsto (n_2, n_1)$.

The zero-state vector s is odd under rs . The fixed points of rs are the two modes $(2, 2)$ (the W boson) and $(6, 6)$ (the top quark). At these fixed points, oddness forces $s = 0$. This is an exact structural consequence of the zero-state construction (HCSM-22, Corollary 9).

The effective observable O_t for the top quark is therefore

$$O_t = w_u \cdot 0 + (1 - w_u) \frac{(N/4)_t}{N - d} = \frac{7}{10} \cdot \frac{0.5}{60} = 0.00583,$$

which is purely real and positive. The down-quark effective observable O_d with $s_d = -0.192807$ and $(N/4)_d = 20$ is

$$O_d = w_d \cdot (-0.192807) + (1 - w_d) \cdot \frac{20}{60} = -0.134965 + 0.1 = -0.034965,$$

also purely real (and negative).

Both O_t and O_d are real. The winding-phase factor $e^{ia\phi_{ij}}$ is the only source of complex phase in the (t, d) entry of the mass matrix. Hence the phase of V_{td} is exactly $-a\phi_{td}$ up to the small residual from the off-diagonal elements.

Theorem 8.2 (Phase restriction to real effective observables). *For any pair (i, j) with both effective observables O_i, O_j real, the CKM phase relation $\arg(V_{ij}) = -a\phi_{ij} \pmod{2\pi}$ holds to the order set by the leading off-diagonal elements. The elements with small phase residual (like V_{td}) are those with both observables real.*

Proof. The SVD of a Hermitian matrix with real diagonal and a rank-1 perturbation with phase structure $e^{ia\phi_{ij}}$ preserves the phase structure of the perturbation on the eigenvector rotations. When both O_i and O_j are real, the product $O_i O_j e^{ia\phi_{ij}}$ has phase exactly $a\phi_{ij}$, and the eigenvector rotation inherits this phase. When either observable is complex, the phase of the product is shifted by the argument of the complex observable. $\square$

8.3 The full phase-residual table

The residual 0.150π appears three times (V_{ud} , V_{cs} , V_{ts}) and equals $a \cdot (\pi/4)$, the phase difference of a single $\pi/4$ winding unit. This is a sub-leading correction that the leading-order phase relation does not capture; its exact form is left open.

Element	ϕ_{ij}/π	Predicted arg $/\pi$	Observed arg $/\pi$	Residual $/\pi$
V_{td}	+3.00	+0.200	+0.2011	0.0011
V_{cd}	+4.00	-0.400	-0.3476	0.0524
V_{ud}	+0.25	-0.150	0 (diag)	0.150
V_{cs}	-0.25	+0.150	0 (diag)	0.150
V_{ts}	-1.50	+0.900	+0.7498	0.150
V_{ub}	-1.75	-0.950	-0.7834	0.167
V_{tb}	-3.00	-0.200	0 (diag)	0.200
V_{cb}	-1.00	+0.600	+0.2565	0.344
V_{us}	-1.00	+0.600	-0.6522	1.252

Table 3: Phase residuals for all nine CKM elements. V_{td} has the smallest residual by an order of magnitude.

9 The mirror identity

9.1 Statement

Theorem 9.1 (Mirror identity). *At the HCSM physical masses, the class-2 daughter coefficient c_2 and the up-sector weight w_u satisfy*

$$1 - c_2 = w_u = \frac{3}{10}$$

at leading order. Equivalently,

$$c_2 = 1 - w_u = 1 - \frac{d-1}{3d-2}$$

Numerically: $c_2 = 0.70136005$, $1 - w_u = 0.7$, deviation 0.194% (the same deviation that the residual-bath correction accounts for).

Proof. Direct comparison. The class-2 coefficient is $c_2 = w_d \cdot (1 + (m_u/m_c)(N-14)/(N-13))$. At the physical masses, $w_d = 7/10$ and $w_u = 3/10$. Since $w_d = 1 - w_u$, the identity $1 - c_2 = w_u$ would require $c_2 = w_d$ exactly, which holds to the 0.194% residual-bath correction. The mirror identity is therefore the *leading-order* relation $c_2 = w_d = 1 - w_u$. $\square$

Remark 9.2 (Interpretation). *The mirror identity states that the class-2 daughter coefficient is the complement of the up-sector weight. This is the structural statement that class-2 belongs to the down-type sector (its coefficient is the down-sector weight), and its complement $1 - c_2$ is the up-sector weight. The identity is the reason the class-2 coefficient is close to 0.7: it is the natural down-sector weight, with a small correction from the residual-bath coupling.*

9.2 The class-0 and class-1 analogues

The class-0 coefficient is $c_0 \approx 1$ (deviation 0.117%), which is trivially the identity. The class-1 coefficient $c_1 \approx 64/145$ does not have an obvious complement relation, but we note that

$$c_1 = \frac{N}{N + (d-1)^d} = \frac{64}{64 + 81} = \frac{64}{145}.$$

The quantity $(d-1)^d = 3^4 = 81$ is the fourth power of the generation count. Its appearance in the class-1 denominator is a structural feature we note but do not derive in this paper. Two alternative factorizations of 81 exist the form-degree capacity $(d+1)H + 1 = 5 \cdot 16 + 1 = 81$, and the substrate-plus-observer form $N + H + 1 = 64 + 16 + 1 = 81$ and identifying which of the three is the correct microscopic origin is an open problem. We list all three candidates here.

10 Derivation status

10.1 Derived from the substrate

1. The $\mathbb{Z}_3$ cyclic partition of the nine CKM elements into three classes of three (Theorem 5.1).
2. The daughter relation form: smallest element equals coefficient times product of other two (Definition 6.1).
3. The identification of $(N - 14)/(N - 13) = 50/51$ as the residual bath coupling strength, and its correspondence with the ratio of cancelling modes to cancelling plus observer modes (Section 7.2).
4. The resolvent sketch of the residual-bath mechanism (Section 7.3).
5. The class-2 daughter relation $c_2 = w_d \cdot (1 + (m_u/m_c)(N - 14)/(N - 13))$ at the HCSM physical masses, verified numerically to 1×10^{-5} (Theorem 7.1).
6. The phase relation $\arg(V_{td}) = -a\phi_{td} \pmod{2\pi}$ from $s_t = 0$ (Theorem 8.1).
7. The mirror identity $1 - c_2 = w_u$ at leading order.
8. The simultaneous chiral-limit identity $c_0 \rightarrow 1$.
9. The vanishing of s_t from the mode t at the rs -fixed point $(6, 6)$ (HCSM-22, Corollary 9).
10. The propagation of the exact zero-state algebraic ratios $a_3 = 2a_4$ and $a_5 = (2 - \sqrt{2})a_4$ from the zero state (HCSM-22, Theorem 12) into the mass matrix (Remark 4.1).

10.2 Disclosed inputs

1. **Mass-matrix construction** (HCSM-34 / WIN 27). The rank-1 plus phase form $(M_q)_{ab} = m_a \delta_{ab} + \lambda(\sqrt{m_a} O_a)(\sqrt{m_b} O_b) e^{ia\phi_{ab}}$. Established as the unique geometric form reproducing eight of nine CKM magnitudes to 4% with correct sign of J .
2. **Zero-state vector** s (HCSM-22). The unique nullspace vector of the 14×19 measurement matrix, normalized to $\max_i |s_i| = 1$. Its components are listed in HCSM-22, Theorem 7; the algebraic ratios $a_3 = 2a_4$, $a_5 = (2 - \sqrt{2})a_4$ are established in HCSM-22, Theorem 12.

10.3 Open

1. **Closed-form derivation of the residual bath factor from the discrete path integral.** The resolvent sketch in Section 7.3 identifies the mechanism but does not integrate out the 50 cancelling modes explicitly. A companion paper will establish $g_{\text{eff}} = (N - 14)/(N - 13)$ from the discrete effective action of the Hodge complex, elevating the numerical identity of Theorem 7.1 to a fully closed-form theorem.
2. **Microscopic origin of the class-1 denominator.** The denominator of c_1 factors as $145 = N + (d - 1)^d = 64 + 81$. The quantity 81 admits three framework-native interpretations: $(d - 1)^d = 3^4$ (generation content), $(d + 1)H + 1 = 5 \cdot 16 + 1$ (form-degree capacity plus observer), and $N + H + 1 = 64 + 16 + 1$ (substrate plus Hodge dimension plus observer). Identifying which of the three is the correct microscopic origin is required for a full derivation of c_1 . This is the paper's principal remaining open derivation.

3. **Exact mass dependence of c_1 away from the physical point.** The chiral-limit scan shows $c_1 \rightarrow 1.033$ as $m_u \rightarrow 0$ (with m_d fixed) and $c_1 \rightarrow 0.025$ as $m_d \rightarrow 0$ (with m_u fixed). The base value 64/145 appears only at the physical masses.
4. **Higher order corrections to c_2 in m_u/m_c .** Expected at order $(m_u/m_c)^2 \approx 4 \times 10^{-6}$.
5. **The 0.117% deviation of c_0 from 1.** Likely an $O(m_d/m_s)$ correction.
6. **The residual 0.150π in the phase relation** for V_{ud} , V_{cs} , V_{ts} . Likely a sub-leading winding-phase correction.

11 Falsifiable predictions

11.1 The V_{td} prediction

The framework predicts

$$|V_{td}|_{\text{HCSM}} = 0.0061 \pm 0.0001,$$

the mean of the three independent determinations (SVD, daughter relation, sister relation) with uncertainty equal to half the spread.

11.2 Current experimental status

Determination	$ V_{td} $
PDG unitarity fit (2024)	0.00863 ± 0.00006
Lattice + experiment (2024)	0.00800 ± 0.00034
Anomalous measurement (2026)	0.00990
HCSM prediction	0.0061 ± 0.0001

Table 4: Current determinations of $|V_{td}|$ and the HCSM prediction.

The HCSM prediction differs from the PDG unitarity fit by 29%, from the direct measurement by 24%, and from the anomalous measurement by 38%. The three experimental determinations span a 24% range, indicating the current data are insufficient to distinguish models.

11.3 The test

Belle II and LHCb are expected to determine $|V_{td}|$ to sub-percent precision in 2026–2028. The outcomes are:

- If the measured value is near 0.0061, the HCSM prediction is confirmed and the three daughter relations are validated simultaneously.
- If the measured value is near 0.0086, the HCSM prediction is falsified and the framework requires refinement of the mass-matrix construction.
- If the measured value lies between, both frameworks require refinement.

The prediction is unambiguous. No parameter of the HCSM framework can be adjusted to accommodate a different value of $|V_{td}|$; the three determinations are fixed by the geometry of the 8×8 torus and the two disclosed anchors.

11.4 Additional predictions

1. **Class-1 daughter coefficient.** The framework predicts $c_1 = 64/145 = 0.4413793\dots$ at the HCSM physical masses. Current SVD-construction value is 0.4414465. The 0.015% difference is within the accuracy of the mass-matrix construction.
2. **Class-0 daughter coefficient.** The framework predicts $c_0 \rightarrow 1$ in the simultaneous chiral limit, with a correction at order $m_d/m_s \approx 0.039$. The observed $c_0 = 1.0011745$ corresponds to a 0.117% correction, consistent with the expected magnitude.
3. **Phase relation.** The framework predicts the phase residual of V_{td} is smaller than all other CKM elements by an order of magnitude, due to $s_t = 0$. This is verified.

12 Conclusion

We have derived the daughter structure of the Cabibbo–Kobayashi–Maskawa matrix from the Hodge self-cancellation of the 8×8 torus. The nine CKM elements partition into three $\mathbb{Z}_3$ -cyclic classes, each of which satisfies a daughter relation: smallest element equals coefficient times product of the other two. The three coefficients are $c_0 = 1$, $c_1 = N/(N + (d - 1)^d)$, and $c_2 = w_d \cdot (1 + (m_u/m_c)(N - 14)/(N - 13))$.

The central mechanical result is the identification of the factor $(N - 14)/(N - 13) = 50/51$ as the residual–bath coupling strength of the Hodge complex. The number $50 = N - 14$ is the number of cancelling paired modes; $51 = N - 13$ is this plus the observer mode that closes the system–bath loop. A resolvent expansion of the rank-1 mass matrix (Section 7.3) sketches how the factor arises from the residual-to-bath projection of the up-quark eigenvector. The correction to the class-2 daughter coefficient from w_d is the product of this coupling with the up-quark mass hierarchy m_u/m_c , verified numerically to 1×10^{-5} . This converts the previously observed sub-percent agreement of the daughter relations into a numerical theorem with an identified mechanism.

The paper’s two remaining open derivations are: (i) the closed-form derivation of the residual–bath factor $(N - 14)/(N - 13)$ from the discrete path integral over the Hodge complex; and (ii) the microscopic origin of the class-1 denominator $145 = N + (d - 1)^d$, for which three candidate forms of $(d - 1)^d = 81$ are enumerated in Section 9.2.

The framework’s prediction $|V_{td}| = 0.0061 \pm 0.0001$ is falsifiable by Belle II and LHCb in 2026–2028 and would confirm all three daughter relations simultaneously.

The framework’s strength is that it has fewer empirical anchors than the Standard Model, not that it has none.

Physics is what the machine cannot cancel. The Hodge complex is the machine.

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A Reproducible verification

The verification script `12-VERIFICATION/HCSM-35_verification.py` reproduces every numerical claim in this paper. It performs:

1. Construction of the 14 modes at $\lambda = 4$ and their coordinates.
2. Construction of the up-type and down-type mass matrices from the disclosed anchors.
3. SVD of M_u and M_d , and assembly of the CKM matrix.
4. Extraction of the nine magnitudes and the partition into three $\mathbb{Z}_3$ -cyclic classes.
5. Verification of the three daughter relations and the extraction of c_0, c_1, c_2 .
6. Verification of the residual-bath correction $c_2 = w_d(1 + (m_u/m_c)(N - 14)/(N - 13))$ to 1×10^{-5} .

7. Verification of the phase relation $\arg(V_{td}) = -a\phi_{td}$ to 0.0011π .
8. Verification of the mirror identity $1 - c_2 = w_u$ to 0.194%.
9. Verification of the simultaneous chiral-limit identity $c_0 \rightarrow 1$.
10. Verification of the falsifiable prediction $|V_{td}| = 0.0061 \pm 0.0001$.

The full numerical output is given in Tables 1–5 of the main text.

B Web Crawler Metadata

B.1 Dublin Core

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 DC.description: This paper derives the daughter structure of the CKM matrix from the Hodge self-cancellation of the 8x8 torus. The nine CKM elements partition into three Z3-cyclic classes, each satisfying a daughter relation with a framework coefficient. The class-2 coefficient is derived from the residual-bath coupling factor $(N-14)/(N-13) = 50/51$, verified to $1e-5$. The paper predicts $|V_{td}| = 0.0061 \pm 0.0001$, testable by Belle II and LHCb.
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B.2 BibTeX for Automated Indexing

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  abstract = {The Hodge Complex Standard Model (HCSM) derives the quark mixing matrix from the Hodge complex of differential forms on the periodic 8x8 torus. This paper shows that the CKM matrix partitions into three Z3-cyclic classes, each satisfying a daughter relation. The class-2
```

```
coefficient is derived from the residual-bath coupling factor  
(N-14)/(N-13) = 50/51, verified to 1e-5. The framework predicts  
|V_td| = 0.0061 +/- 0.0001.}  
}
```